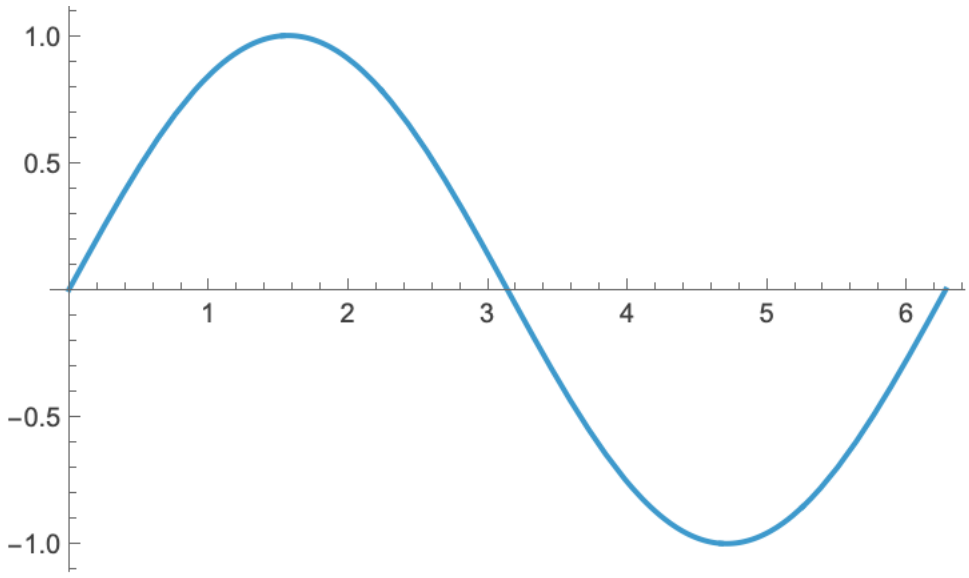
plot

```
Plot[Sin[x], {x, 0, 2 Pi}]
```




```
f[r_] := 1-2*M/r
```

```
f[r]
```

$$1-\frac{2M}{r}$$

```
Series[1 - (2*M)/r,{r,Infinity,2}]
```

$$1-\frac{2M}{r}+O\left(\left(\frac{1}{r}\right)^3\right)$$

```
Integrate[x^2,{x,0,b}]
```

$$\frac{b^3}{3}$$

```
?Series
```

```
InformationData[<|"ObjectType" -> "Symbol", "Usage" -> "\!\(*RowBox[{\"Series\", \"[\", RowBox[{
    StyleBox[\"f\", \"TI\"], \",\", RowBox[{\"{\", RowBox[{StyleBox[\"x\", \"TI\"], \",\",
    SubscriptBox[StyleBox[\"x\", \"TI\"], StyleBox[\"0\", \"TR\"]], \",\", StyleBox[\"n\", \"TI
    \"]}], \"}\"}]}], \"]\"}]\) generates a power series expansion for \!\(*StyleBox[\"f\", \"
```

```
TI\"]\) about the point \!\(*RowBox[{StyleBox[\"x\", \"TI\"], \"=\", SubscriptBox[StyleBox
[\"x\", \"TI\"], StyleBox[\"0\", \"TR\"]]}]\) to order \!\(*SuperscriptBox[RowBox[{\"(\",
RowBox[{StyleBox[\"x\", \"TI\"], \"-\", SubscriptBox[StyleBox[\"x\", \"TI\"], StyleBox
[\"0\", \"TR\"]]}], \")\"}], StyleBox[\"n\", \"TI\"]]\), where \!\(*StyleBox[\"n\", \"TI
\"]\) is an explicit integer. \n\!\(*RowBox[{\"Series\", \"[\", RowBox[{StyleBox[\"f\", \"
TI\"], \",\", RowBox[{StyleBox[\"x\", \"TI\"], StyleBox[\"ï ¢\", \"TI\"], SubscriptBox[
StyleBox[\"x\", \"TI\"], StyleBox[\"0\", \"TR\"]]}]}], \"]\"}]\) generates the leading term
 of a power series expansion for \!\(*StyleBox[\"f\", \"TI\"]\) about the point \!\(*RowBox
[{StyleBox[\"x\", \"TI\"], \"=\", SubscriptBox[StyleBox[\"x\", \"TI\"], StyleBox[\"0\", \"
TR\"]]}]\). \n\!\(*RowBox[{\"Series\", \"[\", RowBox[{StyleBox[\"f\", \"TI\"], \",\",
RowBox[{\"{\", RowBox[{StyleBox[\"x\", \"TI\"], \",\", SubscriptBox[StyleBox[\"x\", \"TI
\"], StyleBox[\"0\", \"TR\"]], \",\", SubscriptBox[StyleBox[\"n\", \"TI\"], StyleBox[\"x\",
 \"TI\"]]}], \"}\"}], \",\", RowBox[{\"{\", RowBox[{StyleBox[\"y\", \"TI\"], \",\",
SubscriptBox[StyleBox[\"y\", \"TI\"], StyleBox[\"0\", \"TR\"]], \",\", SubscriptBox[
StyleBox[\"n\", \"TI\"], StyleBox[\"y\", \"TI\"]]}], \"}\"}], \",\", StyleBox[\"â ¦\", \"TR
\"]}], \"]\"}]\) successively finds series expansions with respect to \!\(*StyleBox[\"x\",
\"TI\"]\), then \!\(*StyleBox[\"y\", \"TI\"]\), etc. ", "Documentation" -> <|"Web" -> "http
://reference.wolfram.com/language/ref/Series.html"|>, "OwnValues" -> None, "UpValues" ->
None, "DownValues" -> None, "SubValues" -> None, "DefaultValues" -> None, "NValues" -> None
, "FormatValues" -> None, "Options" -> {Analytic -> True, Assumptions :> $Assumptions,
SeriesTermGoal -> Automatic}, "Attributes" -> {Protected}, "FullName" -> "System`Series"|>,
 False]
```


```
Integrate[Sin[x]^2, {x, 0, Pi}]
```

$$\frac{\pi}{2}$$